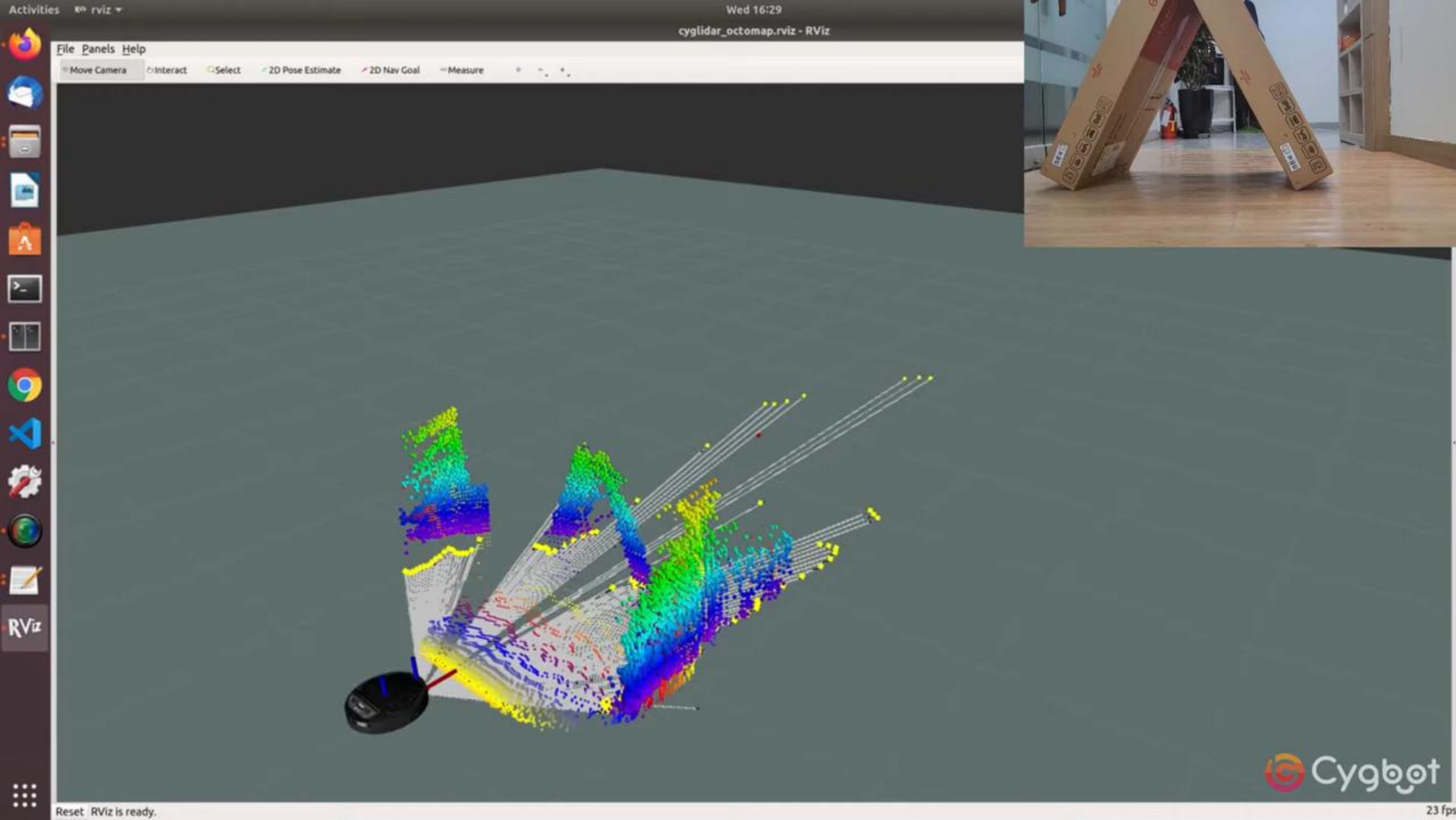


Machine Teaching (and requested topics)

Dylan Losey



This Lecture



- How can robots efficiently gain information from humans?
- Answering some of your questions



How can
robots
efficiently gain
information?



Entropy

Shannon **entropy** measures how *uncertain* we are about a variable.

$$H(X) = - \sum_{x \in X} P(x) \log_2 P(x)$$

$$H(X) = -E[\log_2 P(x)]$$

Entropy

Example. Tossing a two-sided coin with $P(\text{heads}) = 0.5$

$$H(X) = - \sum_{x \in X} P(x) \log_2 P(x)$$

Can you solve for the entropy?

Entropy

Example. Tossing a two-sided coin with $P(\text{heads}) = 0.5$

$$H(X) = - \sum_{n=1}^2 0.5 \log_2 0.5 = -\log_2 0.5 = \mathbf{1}$$

Entropy

Example. Tossing a two-sided coin with $P(\text{heads}) = 0.9$

$$H(X) = - \sum_{x \in X} P(x) \log_2 P(x)$$

Can you solve for the entropy?

Entropy

Example. Tossing a two-sided coin with $P(\text{heads}) = 0.9$

$$H(X) = -0.9 \log_2 0.9 - 0.1 \log_2 0.1 = \mathbf{1/2}$$

If entropy measures **uncertainty**, does it make sense that $P(\text{heads}) = 0.5$ has more entropy than $P(\text{heads}) = 0.9$?

Entropy

Conditional entropy captures uncertainty about x given y .

$$H(X | Y) = \sum_{y \in Y} P(y) \underbrace{H(X | Y = y)}$$

Entropy at a specific y

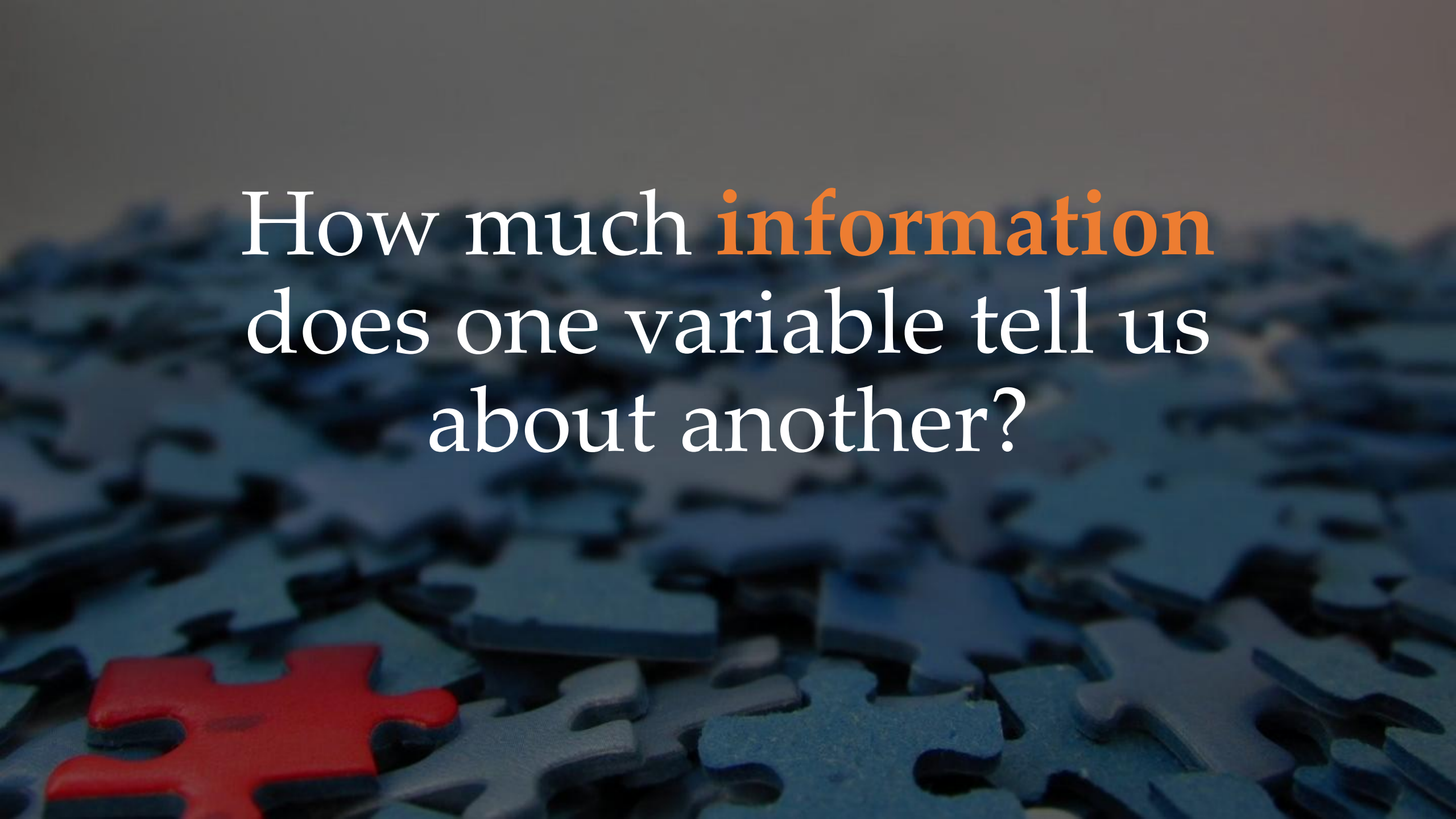
Taken in expectation across Y

Entropy

Conditional entropy captures uncertainty about x given y .

$$H(X | Y) = - \sum_{y \in Y} P(y) \sum_{x \in X} P(x | y) \log_2 P(x | y)$$

$$H(X | Y) = - \sum_{y \in Y} \sum_{x \in X} P(x, y) \log_2 P(x | y)$$



How much **information**
does one variable tell us
about another?

Information Gain

Information gain (*mutual information*) captures the amount of information obtained about x by observing y .

$$I(X; Y) = H(X) - H(X|Y)$$



Uncertainty about x



Uncertainty about x after observing y

Information Gain

Information gain (*mutual information*) captures the amount of information obtained about x by observing y .

$$I(X; Y) = H(X) - H(X|Y)$$

Properties:

- Non-negative. $I(X; Y) \geq 0$
- Symmetric. $I(X; Y) = I(Y; X)$

Information Gain

Conditional information gain (*conditional mutual information*) captures the information obtained about x by observing y given z .

$$I(X; Y | Z) = H(X | Z) - H(X | Y, Z)$$



Uncertainty about x given z



Uncertainty about x given observing y and z

Information Gain

Conditional information gain (*conditional mutual information*) captures the information obtained about x by observing y given z .

$$I(X; Y | Z) = H(X | Z) - H(X | Y, Z)$$

Properties:

- Non-negative. $I(X; Y | Z) \geq 0$
- Symmetric. $I(X; Y | Z) = I(Y; X | Z)$

In **active learning** robots
take actions to gather
information

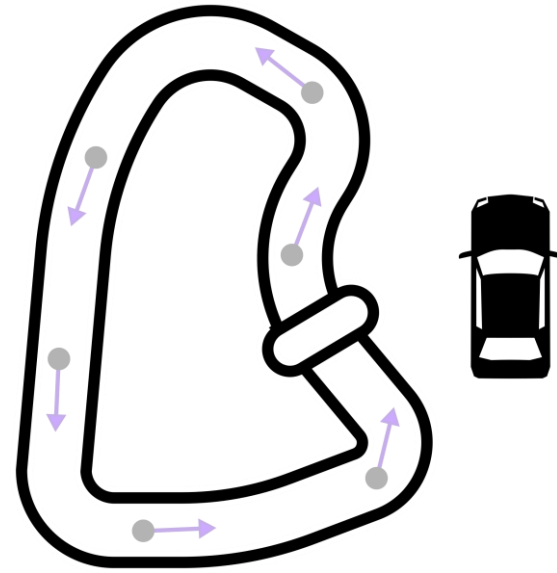


Active Learning

Robots can learn from humans **passively**

- Perform the task optimally based on what the robot knows (exploit)
- Wait for the human to provide feedback

Example. Behavior cloning



Active Learning

Robots can learn from humans **passively**

- Perform the task optimally based on what the robot knows (exploit)
- Wait for the human to provide feedback

But robots can also learn from humans **actively**

- Purposely move to regions where the robot is confused (explore)
- Ask the human questions to remove uncertainty

Active Learning

Formalizing **active learning**

Consider a robot trying to learn θ from the human. The robot can move to any state s , and at s the human provides action a

Let $b^t(\theta)$ be the robot's current belief over θ

Active Learning

Formalizing **active learning**

Consider a robot trying to learn θ from the human. The robot can move to any state s , and at s the human provides action a

Let $b^t(\theta)$ be the robot's current belief over θ

$$b^{t+1}(\theta) \propto P(a \mid s, \theta) b^t(\theta)$$

Model of how human will respond at s

Current estimate of θ

Active Learning

Formalizing **active learning**

Consider a robot trying to learn θ from the human. The robot can move to any state s , and at s the human provides action a

In active learning, choose state s^ that will maximize robot's information gain*

$$s^* = \arg \max_{s \in S} I(\theta; a \mid s, b)$$

Active Learning

Formalizing **active learning**

Consider a robot trying to learn θ from the human. The robot can move to any state s , and at s the human provides action a

In active learning, choose state s^ that will maximize robot's information gain*

$$s^* = \arg \max_{s \in S} \left(H(\theta \mid s, b) - \mathbb{E}_a H(\theta \mid a, s, b) \right)$$

Initial entropy over θ

Entropy over θ after we get the human's response

Active Learning

Formalizing **active learning**

Consider a robot trying to learn θ from the human. The robot can move to any state s , and at s the human provides action a

In active learning, choose state s^ that will maximize robot's information gain*

$$s^* = \arg \max_{s \in S} \left(H(a \mid s, b) - \mathbb{E}_{\theta} H(a \mid s, \theta) \right)$$

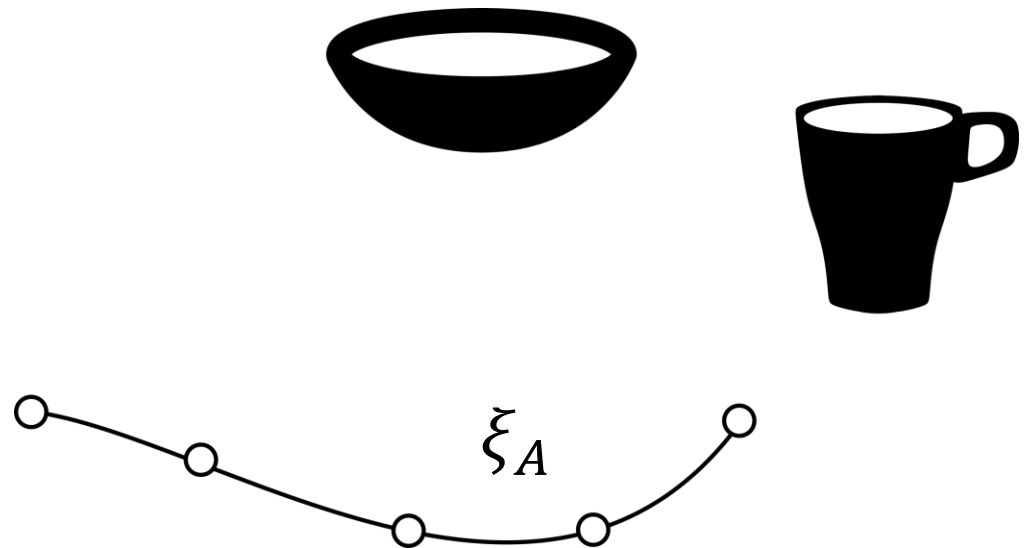
Robot's uncertainty over the human's response

Human's uncertainty over their response

Questions

Robot questions are *multiple choice options*.

The robot shows the human N **motions**, and the human selects the one that best matches their desired behavior.



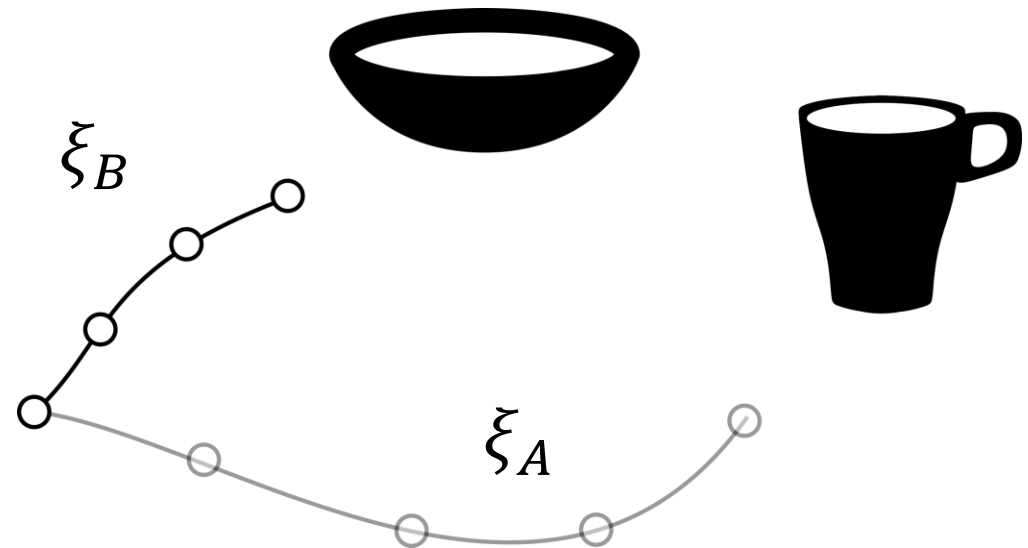
Questions

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$$Q = \{\xi_A, \xi_B\}$$

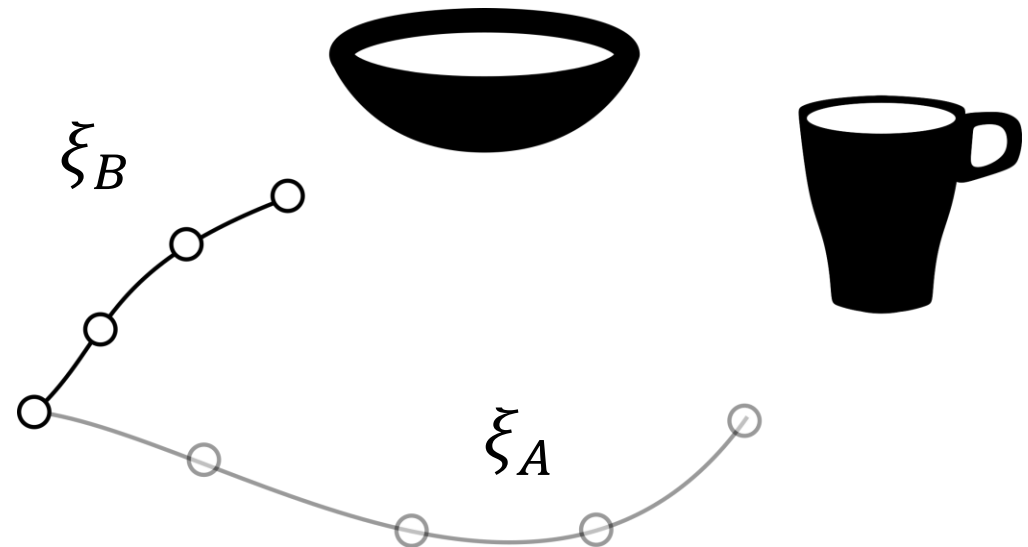
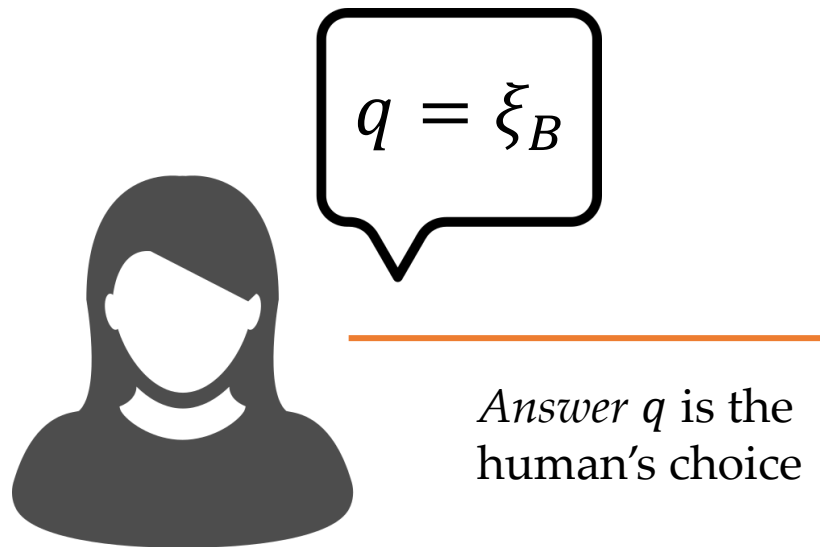
Question is a set of options,
can be more than two.



Questions

Robot questions are *multiple choice options*.

The robot shows the human N **motions**, and the human selects the one that best matches their desired behavior.



Questions

Robot questions are *multiple choice options*.

The robot shows the human N **motions**, and the human selects the one that best matches their desired behavior.

- $Q = \{\xi_A, \xi_B, \dots\}$ is a multiple choice question
- $q = \xi_i$ is the human's answer (their favorite)
- θ are the reward parameters

Questions

Robot questions are *multiple choice options*.

The robot shows the human N **motions**, and the human selects the one that best matches their desired behavior.

$$P(\theta \mid q, Q) \propto P(q \mid Q, \theta) \cdot P(\theta)$$

Our *belief* after seeing
human's answer

$P(q \mid Q, \theta)$ is how likely person answers q to Q
given their reward parameters are θ

Questions

Robot questions are *multiple choice options*.

The robot shows the human N **motions**, and the human selects the one that best matches their desired behavior.

$$P(\theta \mid q, Q) \propto P(q \mid Q, \theta) \cdot P(\theta)$$

When learning from state-action pairs (e.g., *reward learning, shared autonomy*), we used:

$$P(\theta \mid a, s) \propto P(a \mid s, \theta) \cdot P(\theta)$$

We are doing the same thing, the “state” is the **question** and the “action” is the **answer**.

Questions

Robot questions are *multiple choice options*.

The robot shows the human N **motions**, and the human selects the one that best matches their desired behavior.

$$P(\theta \mid q, Q) \propto P(q \mid Q, \theta) \cdot P(\theta)$$

And so we are left with our familiar problem:
how do we *model* the human?

Questions

Assume that the question has two possible answers ξ_A, ξ_B

$$P(q = \xi_i \mid Q, \theta) = \frac{\exp R(\xi_i, \theta)}{\exp R(\xi_A, \theta) + \exp R(\xi_B, \theta)}$$

i is what the human picks,
either $i = A$ or $i = B$

Here $R(\xi, \theta)$ is the total reward
for trajectory ξ given parameters θ

Questions

Assume that the question has two possible answers ξ_A, ξ_B
... or the person can say *IDK* (“I don’t know”)

$$P(q = \xi_i \mid Q, \theta) = \frac{\exp R(\xi_i, \theta)}{\exp R(\xi_i, \theta) + \exp(1 + R(\xi_{-i}, \theta))}$$

$$P(q = idk \mid Q, \theta) = P(q = \xi_A \mid Q, \theta) \cdot P(q = \xi_B \mid Q, \theta) \cdot (\exp(2) - 1)$$

i is what human picks, $-i$ is the other one. Example. If $i = A$, then $-i = B$

Asking Easy Questions: A User-Friendly Approach to Active Reward Learning



What are POMDPs?



POMDP

A Markov decision process (**MDP**) is a tuple:

$$\mathcal{M} = \langle S, A, f, r, \gamma \rangle$$

- $s \in S$ are states

*In MDP, robot knows
exactly what the state is*

POMDP

A partially observable Markov decision process (**POMDP**) is a tuple:

$$\mathcal{M} = \langle S, A, f, r, o, P, \gamma \rangle$$

- $s \in S$ are states

*In POMDP, robot never
directly observes the state*

POMDP

A partially observable Markov decision process (**POMDP**) is a tuple:

$$\mathcal{M} = \langle S, A, f, r, o, P, \gamma \rangle$$

- $s \in S$ are states
- $o \in O$ are observations

Instead of observing the state directly, the robot gets measurements o

POMDP

A partially observable Markov decision process (**POMDP**) is a tuple:

$$\mathcal{M} = \langle S, A, f, r, o, P, \gamma \rangle$$

- $s \in S$ are states
- $o \in O$ are observations
- $P(o \mid s)$ is the likelihood function

*P links the hidden state and
the observed sensor
measurement*

POMDP

In **MDPs**, the robot's policy is a mapping from states to actions:

$$\pi(s) = a$$

*But what happens when we
don't know exactly what the
state is?*

POMDP

In **POMDPs**, the robot's policy is a mapping from **beliefs** to actions:

$$\pi(b) = a$$

- Belief is just another word for the posterior distribution:

$$b(s) = P(s | o^1, \dots, o^t)$$

- Solving POMDPs is harder than solving MDPs, because now the “state” is this continuous probability distribution

POMDP

Consider a **special case** of POMDPs where the state has two components:

$$\mathbf{s} = (s, \theta^*)$$

- s is our usual state (e.g., the position of the robot)
- θ^* is the goal that the human wants to reach

We know the position s but **not** the human's goal θ^*



Other
requested
topics...

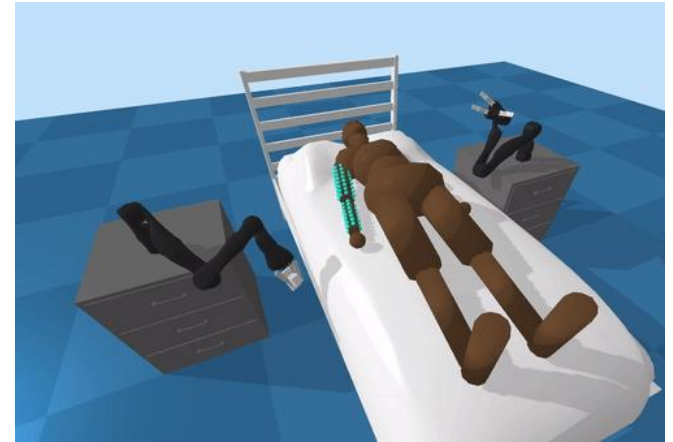


Sim to Real

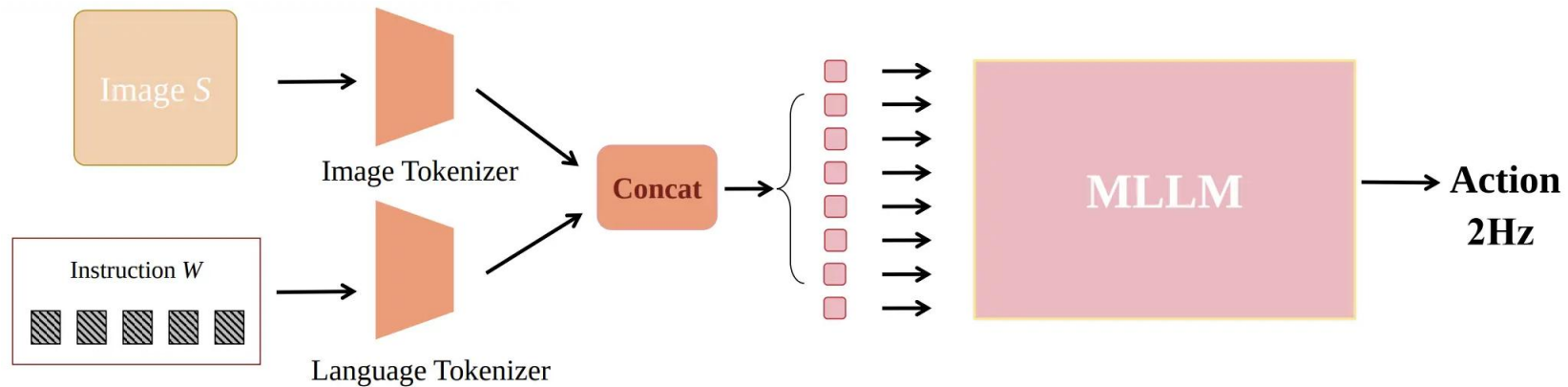
Humans are difficult to simulate; we generally do not use simulation to develop policies that interact well with humans.

Simulation is used to develop autonomous robot policies.

If you want to use simulation to develop an interactive policy (that can work with humans), assume a “worst-case” human within bounds. This goes back to game theory.



VLA_s



Vision-Language-Action (VLA) models are a foundation model for robotics. The idea is to take a state (image + task description) and map that to actions. We often use off-the-shelf image and language tokenizers, and then need to train the VLM/MLLM to map these tokens to actions.

This Lecture



- How can robots efficiently gain information from humans?
- Answering some of your questions

Next Lecture



- Understanding how humans and robots co-adapt.